HireFlow

Smart Recruitment Platform

MERN Stack Edition (MongoDB, Express.js, React, Node.js)

1. Project Overview

HireFlow is a recruitment management platform designed to help companies organize job opportunities, manage candidate applications, and simplify the recruitment process.

Recruitment often involves managing multiple job openings, reviewing numerous CVs, tracking applications, and communicating recruitment decisions. When these activities are handled through emails, spreadsheets, or disconnected tools, recruiters may struggle to maintain an organized overview of candidates and their qualifications.

HireFlow addresses these challenges by providing a centralized platform where recruiters can publish job opportunities, review applications, evaluate candidate qualifications, and track candidates through different recruitment stages.

Candidates can explore available opportunities, submit applications with their CVs, and follow the progress of their applications. They can maintain a default CV in their profile or submit a different CV tailored to each job opportunity.

The platform also integrates artificial intelligence to assist recruiters with CV analysis. AI extracts relevant professional information from submitted CVs, including skills, experience, and education, then compares this information with the requirements of a specific job opening.

The project aims to achieve three main objectives:

- **Simplify recruitment management:** Provide recruiters with tools to publish job openings, organize applications, and manage recruitment workflows.
- **Improve the candidate experience:** Allow candidates to discover opportunities, submit tailored applications, and monitor their application status.
- **Support recruitment analysis with AI:** Help recruiters understand candidate qualifications and identify relevant skills and experience more efficiently.

AI should support recruiters by presenting useful information and explaining how a candidate's qualifications relate to job requirements.

Final recruitment decisions must remain entirely under human control. The platform must not automatically reject candidates or make hiring decisions based on AI-generated assessments.

The final application should provide a complete recruitment workflow, from job publication and candidate application to CV analysis and recruitment status management.

2. Users and Roles

HireFlow supports two main user roles.

Recruiter

Recruiters represent companies or organizations looking to hire candidates. They should be able to:

- Create and manage their recruitment account and basic company information.
- Publish and manage job openings.
- Define job requirements, qualifications, and required skills.
- View applications submitted for their job opportunities.

- Access candidate profiles and the CVs submitted for their positions.
- Use AI to analyze CVs and compare qualifications with job requirements.
- Organize candidates through different recruitment stages.
- Update application statuses and review recruitment activity.

Recruiters should only manage job openings and applications belonging to their recruitment account or organization.

Candidate

Candidates are individuals searching for employment opportunities. They should be able to:

- Create an account and manage their personal profile.
- Maintain a default CV in their profile.
- Browse available job opportunities.
- View job descriptions and required qualifications.
- Apply using their default CV or upload a different CV for a specific opportunity.
- View their submitted applications and associated CVs.
- Follow application statuses as recruiters update them.

Candidates should only access their own personal information and application records.

Public job listings may be accessible without authentication, but candidates must have an account to submit and track applications.

3. Functional Requirements

3.1 Must Have: Required Features

The following features are essential for HireFlow to function as a complete recruitment management platform.

A. Authentication and User Management

The platform must provide secure authentication and role-based access control.

- Recruiters and candidates must be able to register and log in using email and password.
- Users should be able to view and update their basic account information.
- Recruiters should be able to maintain basic company information associated with their job openings.
- Candidates should be able to maintain their personal and professional profiles.
- Users must only access resources and actions permitted by their roles.
- Candidate information and uploaded documents must be protected from unauthorized access.

B. Job Posting Management

Recruiters must be able to create and manage job opportunities. Each job posting should contain relevant information, such as:

Information	Description
Job Title	Position being offered
Company	Organization publishing the opportunity
Location	City, country, or remote work arrangement
Employment Type	Full-time, part-time, internship, contract, or another relevant type
Description	Responsibilities and general information about the position
Required Skills	Technical or professional skills expected from candidates
Qualifications	Relevant education, experience, or other requirements
Application Status	Whether the opportunity is open or closed to applications

The application should allow recruiters to:

- Create, edit, and delete job postings.
- Save job postings as drafts before publication.
- Publish opportunities for candidates to discover.
- Close job openings when applications are no longer being accepted.
- View applications associated with each job posting.
- Access previously published or closed opportunities.

Only published and open job opportunities should accept new applications.

C. Job Discovery and Search

Candidates must be able to discover available job opportunities. The platform should allow candidates to:

- Browse published job openings.
- Search for opportunities using job titles or relevant keywords.
- Filter opportunities by criteria such as location or employment type.
- View complete job descriptions and requirements.
- Access the application process for an available position.

Job listings should present enough information for candidates to understand the position before applying.

D. Candidate Profile and CV Management

Candidates must be able to maintain their professional profiles and manage the CVs used for job applications. The platform should allow candidates to:

- Manage their personal contact information.
- Provide basic professional information.
- Upload and manage a default CV in their profile.
- Use their default CV or upload a different CV when applying for a job.

- View the CV associated with each submitted application.
- Update or replace their default CV without affecting previous applications.

Each application must preserve the exact CV submitted for that position, independently of the candidate's current profile or default CV.

For example, a candidate applying for a Frontend Developer position may submit a CV emphasizing React and Next.js experience, while using a different CV for a Backend Developer position.

The platform should support common document formats, such as PDF, and securely store uploaded documents. Recruiters should only access CVs associated with applications submitted to their job openings.

E. Job Applications

HireFlow must support a complete job application process. Candidates should be able to:

- Apply to an open job opportunity.
- Select their default CV or upload a different CV specifically for the application.
- Include an optional short introduction or motivation message.
- Review their application information before submission.
- Receive confirmation that their application was submitted.
- View their application history.
- Access the CV submitted for each application.
- Follow the current status of each application.

Recruiters should be able to:

- View all applications submitted to their job openings.
- Access candidate information and the CV submitted for each position.
- Review individual applications.
- Update application statuses.
- Organize applicants according to their recruitment stage.

The platform should prevent unintended duplicate applications to the same job opening. Submitted applications must retain their associated CVs even if candidates later update their profiles or default documents.

F. Recruitment Workflow Management

Recruiters must be able to track applications through a clear recruitment workflow. The application should support stages such as:

Recruitment Stage	Description
Applied	The candidate has successfully submitted an application.
Under Review	The recruiter is reviewing the application.
Shortlisted	The candidate has been selected for further consideration.

Recruitment Stage	Description
Interview	The candidate has reached the interview stage.
Hired	The recruiter has completed the recruitment process successfully.
Rejected	The candidate has not been selected for the position.

Recruiters should be able to move applications between appropriate stages and review their current status. Candidates should be able to view the latest status of their applications.

Recruitment status changes must be controlled by authorized recruiters. AI must not automatically advance, reject, or hire candidates.

G. AI-Assisted CV Analysis and Qualification Matching

HireFlow must integrate AI to help recruiters analyze candidate qualifications. The application should:

- Analyze the specific CV submitted for the selected job application.
- Extract relevant professional information, including skills, education, and experience.
- Analyze the requirements of the corresponding job opening.
- Compare extracted candidate information with the job requirements.
- Highlight qualifications that correspond to the position.
- Identify requirements that are not clearly demonstrated in the submitted CV.
- Generate a readable analysis explaining the relationship between the candidate's qualifications and the position.
- Present a job-specific overall qualification match score from 0 to 100 alongside the original application, matching qualifications, unverified requirements, and a short evidence-based explanation for recruiter review.

The analysis should focus on professional qualifications and provide explanations supported by the submitted CV.

For example, if a position requires Next.js, TypeScript, and PostgreSQL experience, the AI should identify which of these skills appear in the candidate's CV and highlight any requirements for which evidence is missing. The AI should not assume that a candidate lacks a skill simply because it is not mentioned in the CV.

The AI analysis must be associated with the specific job application and its submitted CV, rather than the candidate's default profile CV. This ensures that recruiters receive an analysis based on the document the candidate intentionally submitted for their position.

3.2 Should Have: Recommended Features

These features improve recruitment organization and the experience of recruiters and candidates.

A. Candidate Profile Summaries

The platform should provide concise AI-generated summaries of candidate profiles based on the CVs submitted for specific applications. These summaries may highlight:

- Main professional skills.

- Relevant work experience.
- Educational background.
- Qualifications related to the selected position.

Summaries should remain factual and based on information provided by candidates.

B. Application Statistics

Recruiters should have access to an overview of their recruitment activity, including:

- Number of published job openings.
- Total applications received.
- Applications received for each position.
- Number of candidates in each recruitment stage.
- Recent recruitment activity.

These statistics should help recruiters organize their work and understand the current state of their recruitment process.

C. Recruitment Notes

Recruiters should be able to add private notes to individual applications. Notes may contain interview observations, questions for follow-up, or other relevant recruitment information. These notes should only be accessible to authorized recruiters and should not appear in the candidate's application view.

D. Application Status Notifications

Candidates should receive notifications when the status of their application changes. Notifications may be delivered within the application or through email. The objective is to keep candidates informed without requiring them to repeatedly check their application history.

3.3 Could Have: Optional Features

The following improvements offer opportunities for creativity and additional technical exploration.

- **AI CV Improvement Suggestions:** Allow candidates to request suggestions for improving the clarity, organization, and presentation of their CVs.
- **Advanced Application Filters:** Help recruiters search and organize applications using relevant skills, experience, or recruitment stages.
- **Recruitment Analytics:** Provide additional insights into application trends and recruitment activity over time.
- **Interview Scheduling:** Allow recruiters to propose interview dates and communicate them to candidates.
- **Job Recommendations:** Suggest potentially relevant open positions to candidates based on their stated skills and experience.

Students are encouraged to propose other enhancements that improve the recruitment experience or help recruiters organize their work.

Additional features should remain focused on recruitment management rather than expanding HireFlow into a complete enterprise human resources system.

4. AI Integration

4.1 Purpose

The primary purpose of AI in HireFlow is to help recruiters analyze CVs and understand how candidate qualifications relate to the requirements of specific job openings.

Reviewing large numbers of CVs can be repetitive and time-consuming, particularly when professional information is presented in different formats and structures.

HireFlow uses AI to extract relevant information from CVs, organize candidate qualifications, and highlight relationships between those qualifications and job requirements. The AI integration should help recruiters access useful information more efficiently while preserving human control over the recruitment process.

4.2 Expected AI Workflow

The AI functionality should support the following workflow:

- **1. Job Creation:** A recruiter publishes a job opening containing responsibilities, required skills, and relevant qualifications.
- **2. Candidate Application:** A candidate selects their default CV or uploads a tailored CV for the position and submits their application.
- **3. Document Processing:** The application processes the CV associated with that specific application and extracts its readable content.
- **4. Information Extraction:** The AI identifies relevant professional information, including skills, experience, education, and qualifications.
- **5. Job Requirement Analysis:** The AI examines the requirements associated with the selected job opening.
- **6. Qualification Matching:** The AI identifies relationships between the candidate's documented qualifications and the position's requirements.
- **7. Analysis Generation:** The platform presents a structured summary of relevant qualifications, demonstrated strengths, and requirements that could not be confirmed from the CV.
- **8. Recruiter Review:** The recruiter reviews the submitted CV and AI-generated analysis before making any recruitment decisions.

AI analysis should always be associated with the specific application, its submitted CV, and the corresponding job opening. The same candidate may receive different qualification analyses for different positions because both the job requirements and submitted CVs may vary.

4.3 Expected AI Capabilities

The AI integration should be capable of:

- Extracting relevant information from CV documents.
- Identifying technical and professional skills.
- Recognizing educational qualifications and professional experience.
- Understanding job descriptions and required qualifications.

- Comparing candidate information with specific job requirements.
- Highlighting qualifications supported by the submitted CV.
- Identifying requirements that are missing or unclear in the provided information.
- Generating concise qualification summaries and an overall job match score from 0 to 100, supported by the CV evidence and specific job requirements.
- Producing structured results that the application can validate and present.

For example, a company publishes a Full Stack Developer position requiring experience with Next.js, TypeScript, Express.js, and PostgreSQL. A candidate submits a CV describing previous projects using Next.js, TypeScript, and Express.js, but does not mention PostgreSQL.

The AI should identify the relevant technologies demonstrated in the CV and highlight that PostgreSQL experience could not be confirmed. The AI may indicate that this requirement requires further clarification, but should not automatically classify the candidate as unqualified.

The recruiter remains responsible for evaluating the application and deciding whether to move the candidate to the next recruitment stage.

(Note: the technologies named in this example describe a candidate's CV and a job posting. They illustrate the AI behavior only and are not part of the project's technology stack.)

4.4 Technical Considerations

The AI functionality must be implemented within the Express.js backend, as a dedicated AI service module (for example, a separate layer that processes the CV text, builds the prompt, calls the chosen AI API or model, and validates the response). No separate AI backend is required.

Students should consider the following constraints:

- CVs may contain different layouts, formatting styles, and document structures.
- Some documents may contain incomplete information or text that is difficult to extract.
- Extracted information must be validated before being presented as factual.
- AI should not invent qualifications, experience, or educational credentials.
- Missing information should not automatically be interpreted as a missing qualification.
- The job match score must reflect documented job-related qualifications, use a consistent and explainable approach, and be accompanied by evidence, matching skills, and unverified requirements. A missing mention in a CV must not be presented as proof that a candidate lacks the skill.
- The system should avoid evaluating candidates using sensitive personal characteristics unrelated to job requirements.
- Uploaded CVs contain personal information and must be handled securely.
- AI must not automatically reject, shortlist, rank, or hire candidates.
- Recruiters must be able to review applications manually when AI processing fails or is unavailable.
- The AI solution should be achievable using free APIs, open-source models, or suitable free-tier services.

Students may explore document text extraction (using suitable Node.js libraries), natural language processing, and large language models to implement CV analysis and qualification matching.

Complex recruitment scoring algorithms, automated candidate rankings, and enterprise-level applicant screening infrastructure are not required. The match score is an informational aid for review, not a hiring decision or an automatic mechanism to reject, shortlist, or rank candidates.

5. Technology Stack and Deployment

5.1 Required Technology Stack (MERN)

The project must be built exclusively with the MERN stack. No other framework, language, or database is required or expected.

Component	Technology
Database	MongoDB (with Mongoose as the ODM)
Backend	Node.js with Express.js (main application logic and AI integration)
Frontend	React.js (functional components and hooks, with React Router for navigation)
AI Functionality	CV information extraction and job qualification matching, integrated through the Express.js backend using an AI API or model
Version Control	Git and GitHub

MongoDB is well suited to HireFlow, particularly because of its flexible document structures for candidate profiles, CV analysis results, and job requirements. Users, company profiles, job postings, applications, and recruitment stages can be stored in separate collections linked by references. The selected data model should support reliable data relationships, consistent application records, and appropriate access control.

The Express.js backend is responsible for the main application logic, authentication, authorization, recruitment workflows, database operations, and AI-related CV analysis and qualification matching. The React frontend consumes the backend REST API.

Students are responsible for designing the database, organizing application architecture, selecting suitable document storage solutions (for example, MongoDB GridFS or a cloud storage service), and structuring the AI module within the backend.

5.2 Deployment

The completed HireFlow application must be deployed and accessible online. All components, including the frontend, Express.js backend, database, and AI integration, must communicate correctly in the deployed environment.

Suggested deployment services include:

Component	Suggested Service
React Frontend	Vercel or Netlify
Express.js Backend (including AI integration)	Railway or Render

Component	Suggested Service
MongoDB Database	MongoDB Atlas

These providers are suggestions rather than mandatory choices. Students may select alternative hosting services that satisfy the project requirements, preferably using available free tiers or free resources.

The deployed application must support the complete recruitment workflow, including CV uploads and AI-assisted qualification analysis.

6. Expected Deliverables

The final project submission should include:

- A functional full-stack HireFlow application built with the MERN stack.
- Secure authentication and role-based authorization for recruiters and candidates.
- Recruiter and candidate profile management.
- Job posting creation, publication, and management.
- Job discovery and search functionality.
- Candidate applications with default or application-specific CV uploads.
- Independent preservation of the CV submitted for each application.
- Recruitment stage and application status management.
- AI-assisted CV information extraction.
- AI-generated qualification matching against job requirements.
- Recruiter access to applications and AI-generated analyses.
- Candidate application history and status tracking.
- Persistent database integration using MongoDB.
- Source code available in a GitHub repository.
- A README containing the project overview, installation steps, configuration, and deployment instructions.
- Sample job postings, candidate profiles, applications, and CV documents for demonstration.
- A live deployed application accessible online.
- A final presentation and demonstration of the application's main workflows and technical decisions.

Example End-to-End Scenario

- **1. Initial Setup & Job Publication:** Atlas Digital, a technology company, wants to recruit a Junior Full Stack Developer. A recruiter creates a HireFlow account, completes the company profile, and publishes a job opening requiring knowledge of Next.js, TypeScript, Express.js, and PostgreSQL.

- **2. Job Discovery:** The job opportunity becomes visible on the platform. Sara, a junior developer searching for employment, creates a candidate account, completes her professional profile, and uploads a default CV focused on frontend development experience.
- **3. Tailored Application Submission:** Sara discovers the Atlas Digital Full Stack Developer position. Recognizing the need for both frontend and backend skills, she uploads a different CV emphasizing her full-stack projects rather than using her default CV, and submits her application.
- **4. Document & Application Preservation:** The platform saves the application and preserves the submitted CV independently of Sara's default CV profile.
- **5. AI Information Extraction:** The recruiter receives the application and opens Sara's profile. HireFlow processes the submitted CV for the Full Stack Developer position and uses AI to extract relevant skills, education, and project experience.
- **6. Qualification Matching & Gap Analysis:** The AI compares Sara's submitted CV against the job requirements and displays an illustrative overall match score of 78/100, with an explanation of the documented matches. It identifies Next.js, TypeScript, and Express.js experience, while noting that PostgreSQL experience cannot be confirmed from the CV. The score is not a final judgment of her suitability.
- **7. Recruiter Review & Stage Advancement:** The recruiter evaluates both the AI-generated analysis and Sara's submitted CV, then moves her application status from "Applied" to "Shortlisted".
- **8. Candidate Status Tracking:** Sara logs into HireFlow and views her updated application status.
- **9. Profile Update Independence:** A few days later, Sara updates her default profile CV with new project experience. Her previously submitted application to Atlas Digital and its associated CV remain unchanged.
- **10. Interview Invitation:** The recruiter updates Sara's status to "Interview" after deciding to invite her for further evaluation.

Throughout the process, AI assists with understanding the application, while the recruiter remains responsible for all recruitment decisions. This scenario demonstrates the expected connection between job posting management, candidate-specific CV submissions, AI qualification matching, application history, and recruitment workflow tracking.

7. Evaluation Criteria

The project will be evaluated according to the following criteria:

Criterion	Description
Functional Completeness	Correct implementation of the required features and main recruitment workflows.
Frontend and User Experience	Interface quality, responsiveness, ease of use, and clarity of recruiter and candidate interactions.
Backend and Database	Quality of API development, data management, authentication, authorization, and recruitment business logic.

Criterion	Description
AI Integration	Accuracy, reliability, transparency, and practical usefulness of CV information extraction, evidence-based qualification matching, and explainable job-specific match scores.
Code Quality	Organization, readability, maintainability, and appropriate technical decisions.
Deployment	Accessibility and correct integration of all application components, including document processing and AI functionality.
Team Collaboration	Effective collaboration, Git usage, individual contributions, and understanding of the implemented solution.
Final Demonstration	Ability to demonstrate complete workflows, explain technical choices, and present a functional application.

The expected outcome is a coherent, functional, and practical recruitment management platform that demonstrates the ability to combine full-stack MERN development and artificial intelligence to simplify application management, improve the recruitment experience, and support informed human recruitment decisions.